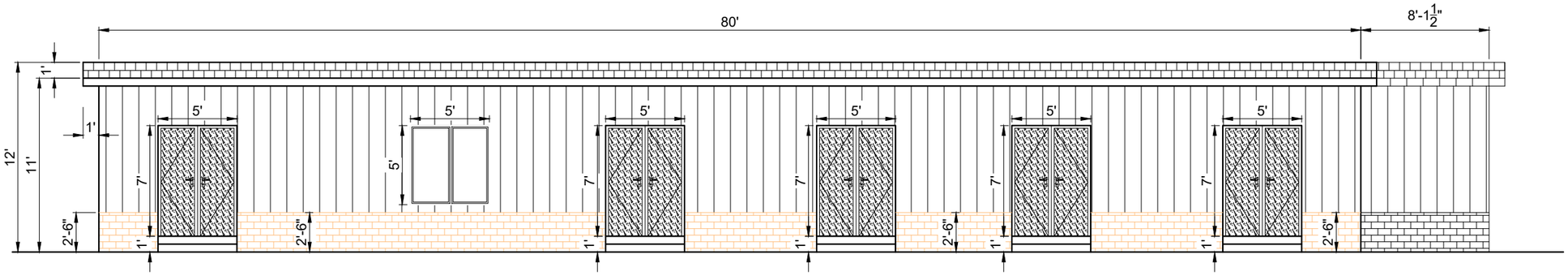
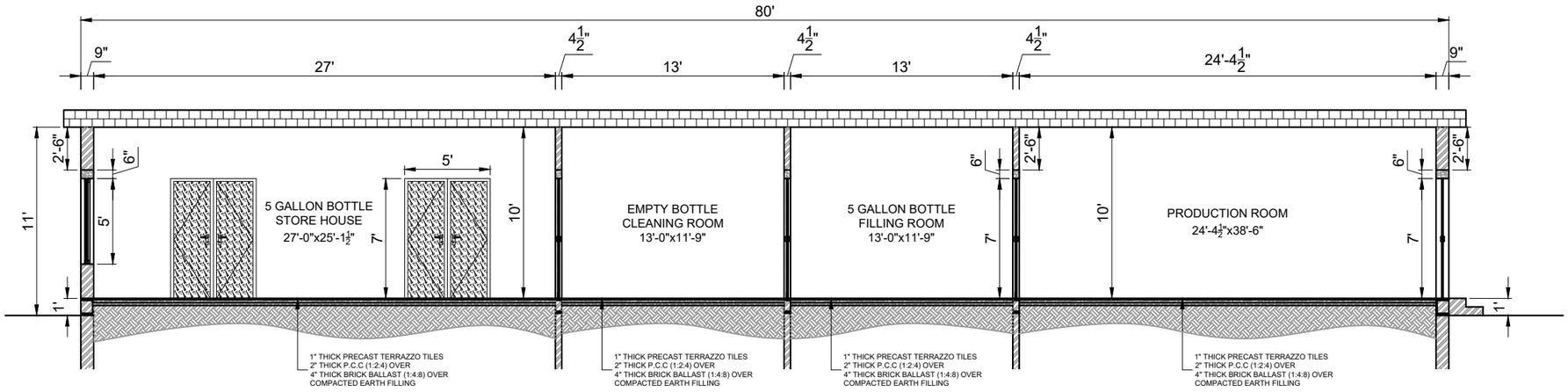


FRONT ELEVATION

SCALE:- 1/8" = 1'-0"



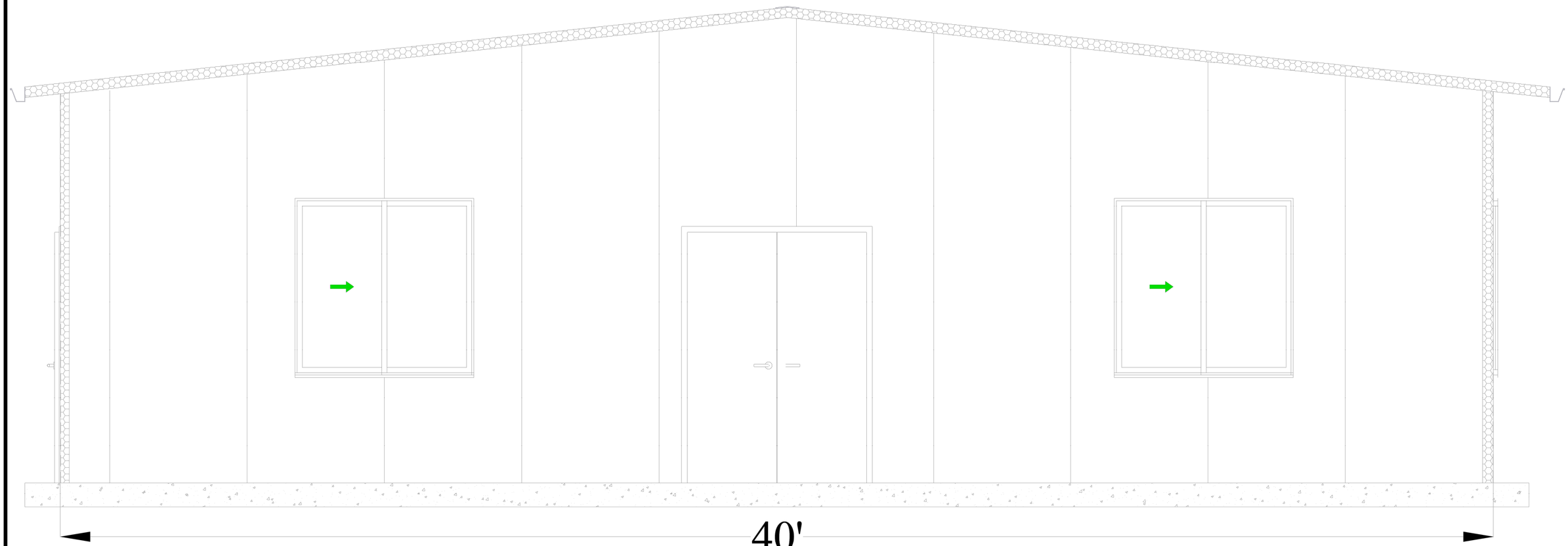
SECTION A-A

SCALE:- 1/8" = 1'-0"

Engineering Consultancy Services Punjab(Pvt.)Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** WATER BOTTLING PLANT (5000LPH) FRONT ELEVATION	**DRAWING NO:** ARCH-02	
			DESIGNED BY:	ENGR. AZIZ				
			CHECKED BY:	ENGR. KAMRAN				
			APPROVED BY:	ENGR. KAMRAN			**SCALE:**	**SHEET:** A3
			DATE:	OCTOBER, 2025				

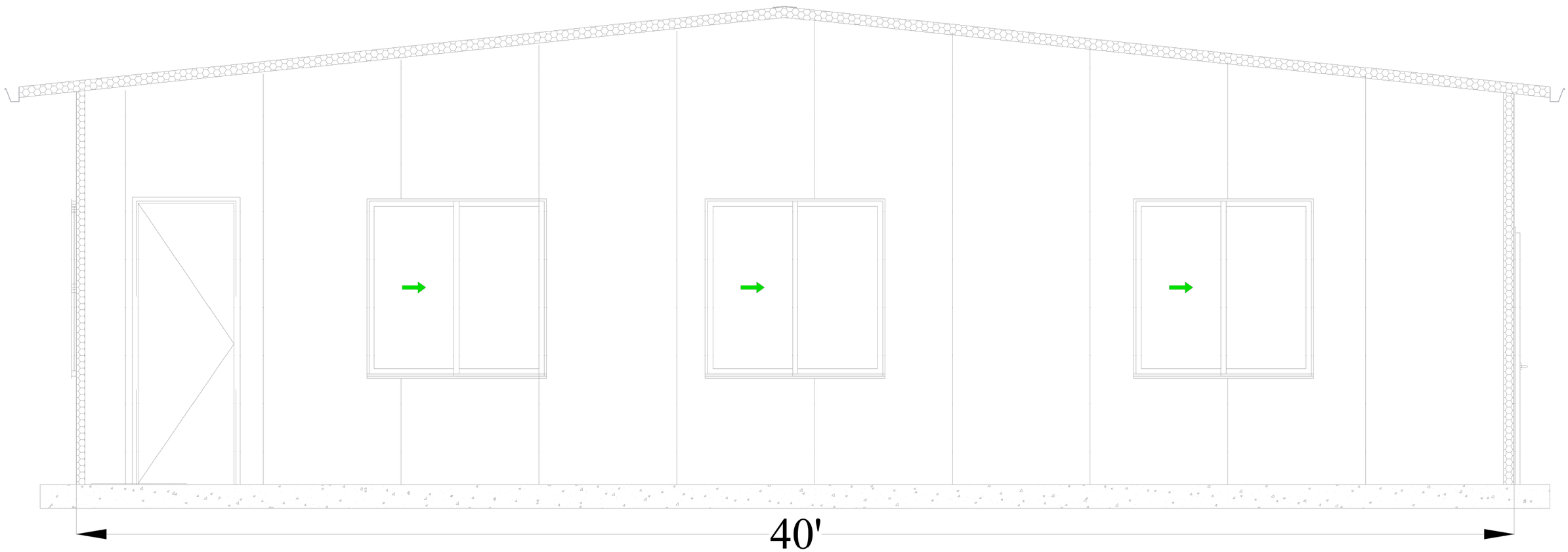
REV.	DESCRIPTION	SIGN.	DATE



RIGHT SIDE

Engineering Consultancy Services Punjab (Pvt.) Ltd. ECSP

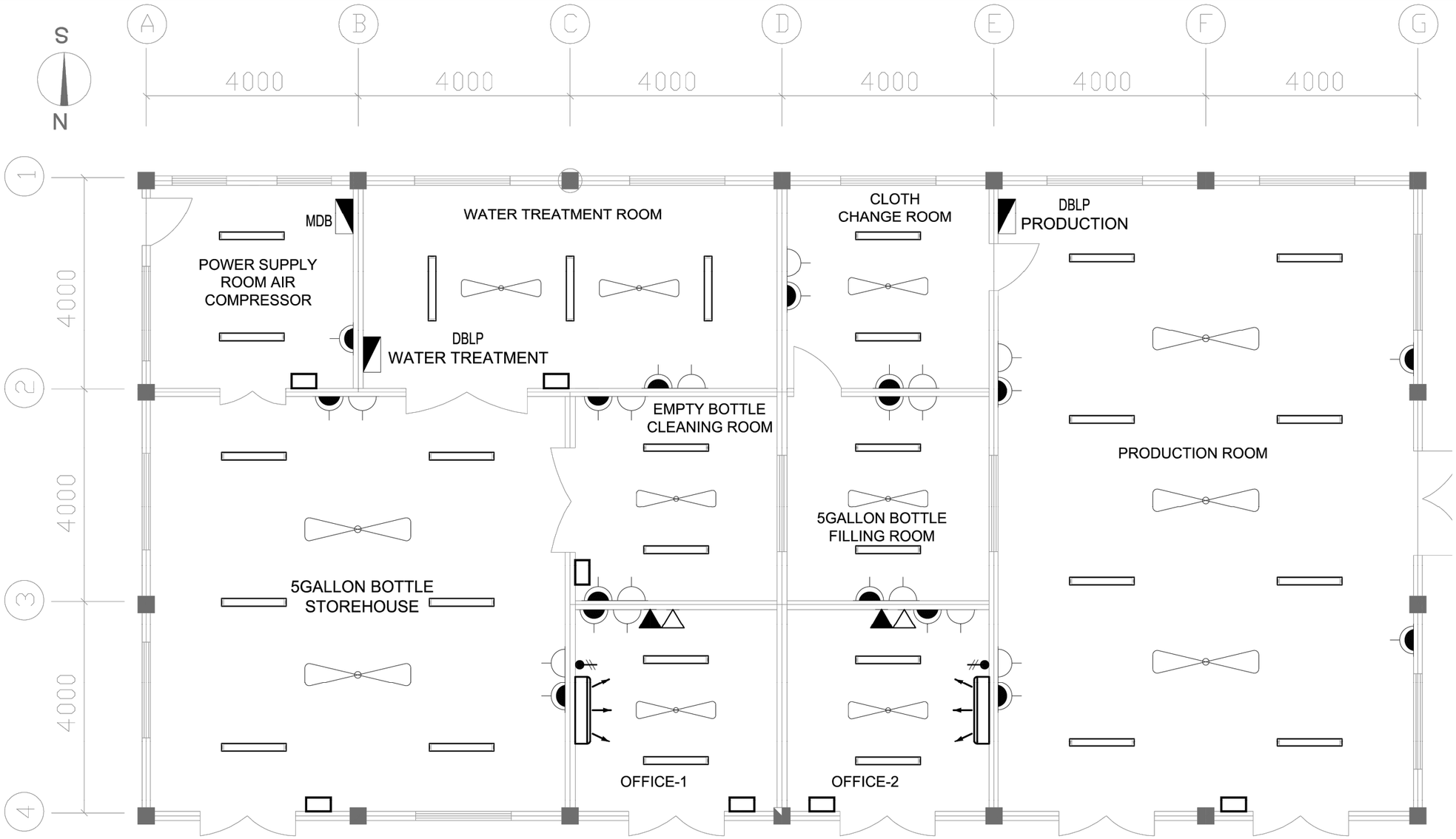
CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY: USMAN MUSTAFA DESIGNED BY: ENGR. AZIZ CHECKED BY: ENGR. KAMRAN APPROVED BY: ENGR. KAMRAN DATE: OCTOBER, 2025	**DRAWING TITLE:** WATER BOTTLING PLANT (5000LPH) RIGHT SIDE ELEVATION	**DRAWING NO:** ARCH-03 **SCALE:** **SHEET:** A3	REV. / DESCRIPTION / SIGN. / DATE



LEFT SIDE

Engineering Consultancy Services Punjab(Pvt.)Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY: USMAN MUSTAFA DESIGNED BY: ENGR. AZIZ CHECKED BY: ENGR. KAMRAN APPROVED BY: ENGR. KAMRAN DATE: OCTOBER, 2025	**DRAWING TITLE:** WATER BOTTLING PLANT (5000LPH) LEFT SIDE ELEVATION	**DRAWING NO:** ARCH-04 **SCALE:** **SHEET:** A3	REV. / DESCRIPTION / SIGN. / DATE		



LEGEND

S.N	DESCRIPTION	SYMBOLS	MOUNTING HEIGHT	S.N	DESCRIPTION	SYMBOLS	MOUNTING HEIGHT
1	RECESSED PVC CONDUIT SIZE 1" DIA	——	RECESSED IN SLAB	6	WIDE BATTEN LIGHT SURFACE / CEILING MOUNTED 36W	▭	----------
2	MAIN DISTRIBUTION BOX (MDB)	⋈	ON WALL AT 1200 F.F.L	7	MULTI STANDARD UNIVERSAL SOCKET 13 AMPS	◠	ON WALL AT 600 F.F.L
3	DISTRIBUTION BOX (DB)	◥	ON WALL AT 1200 F.F.L	8	MULTI STANDARD UNIVERSAL SOCKET 15 AMPS	◠	ON WALL AT 600 F.F.L
4	CEILING FAN	⋈	MOUNTED	9	TELEPHONE OUTLET	◀	----------
5	5A, 220V, ONE WAY, MULTI GANG SWITCH WITH REGULATOR	▭	H=1200/AFFL	10	DATA OUTLET	◁	----------
				11	MULTI STANDARD UNIVERSAL SOCKET 15 AMPS (FOR AC)	▮	----------
				12	AC EVAPORATOR UNIT	▭	WALL MOUNTED

Engineering Consultancy Services Punjab (Pvt.) Ltd. ECSP

CLIENT:
PUNJAB SAAF PANI AUTHORITY

CONSULTANTS:
ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP).
83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan.
+92-42-99333976, +92-42-5717681-4
info@ecsp.com.pk www.ecsp.com.pk

PROJECT:
SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB

DRAWN BY:	USMAN MUSTAFA	
DESIGNED BY:	ENGR. AZIZ	
CHECKED BY:	ENGR. KAMRAN	
APPROVED BY:	ENGR. KAMRAN	
DATE:	OCTOBER, 2025	

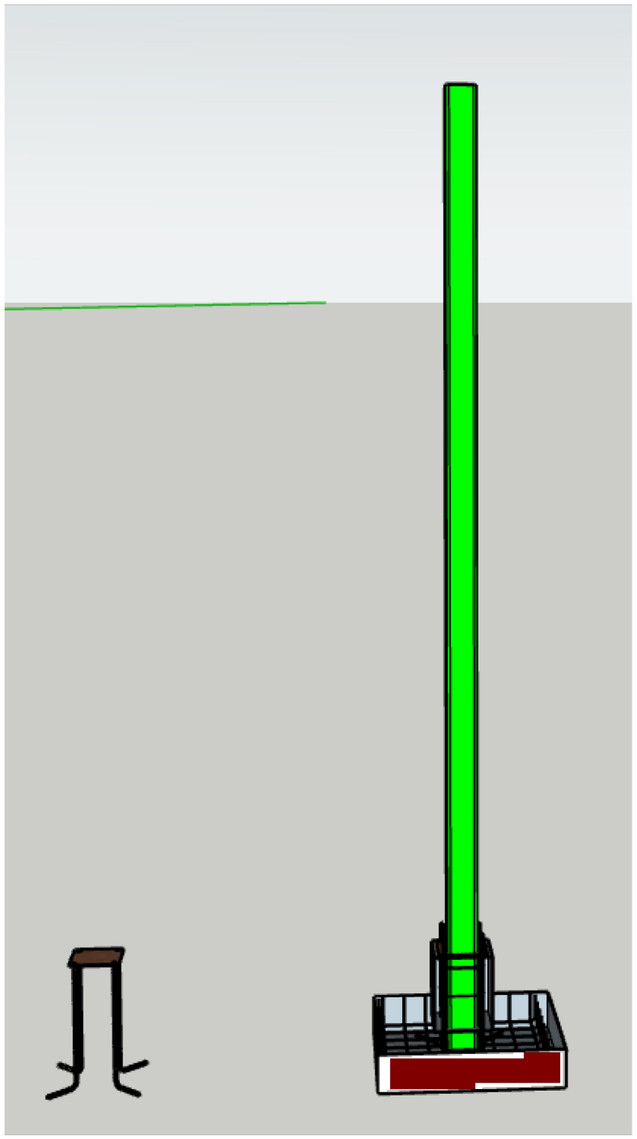
DRAWING TITLE:
WATER BOTTLING PLANT (5000LPH) ELECTRIFICATION LAYOUT PLAN

DRAWING NO: ARCH-05

SCALE:

SHEET: A3

REV.	DESCRIPTION	SIGN.	DATE

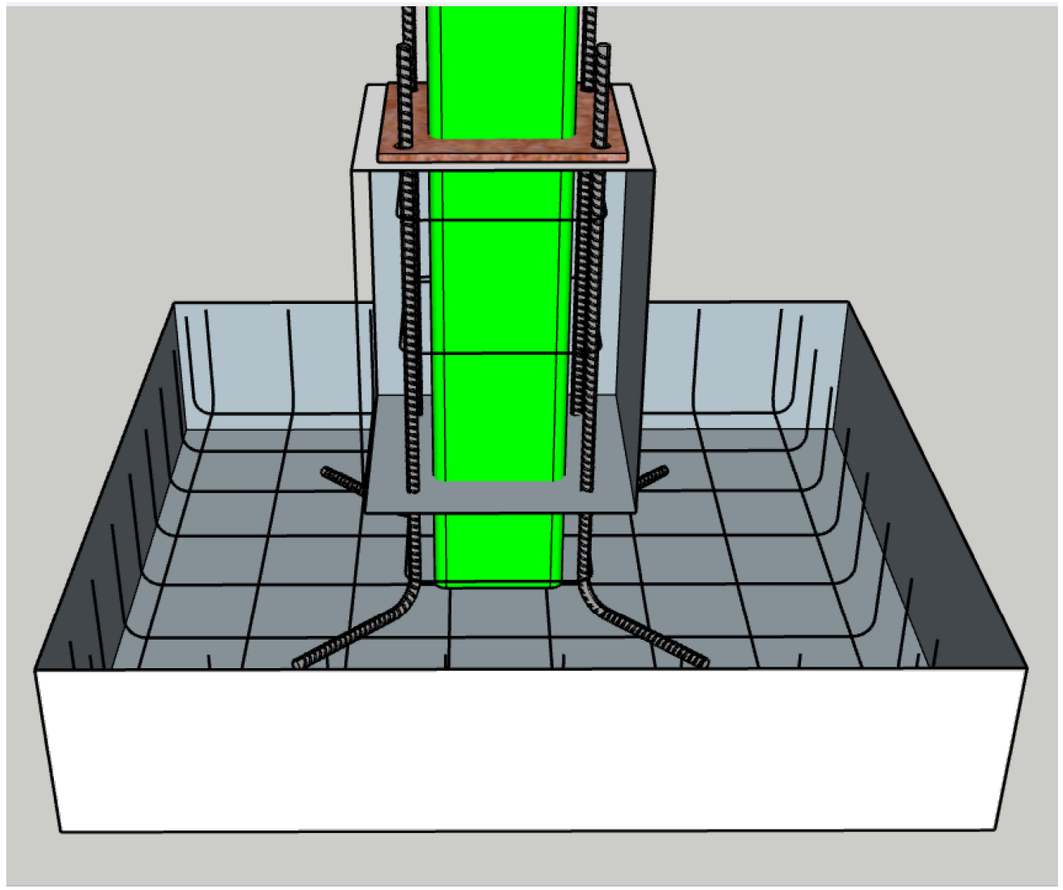


Detail Drawing

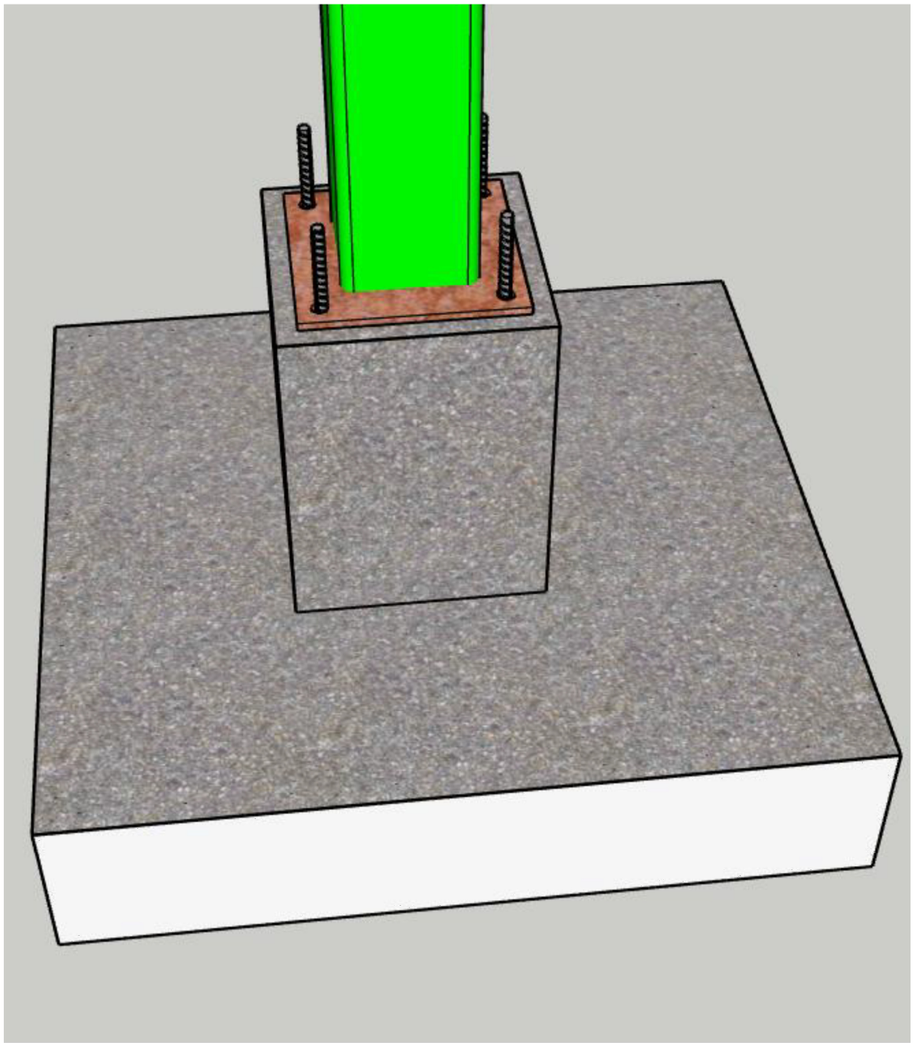
Foundation Plate With J Bolt

Engineering Consultancy Services Punjab (Pvt.) Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** FOUNDATION DETAIL AND FOUNDATION PLATE WITH J BOLT	**DRAWING NO:** ARCH-06		REV.	DESCRIPTION	SIGN.	DATE
			DESIGNED BY:	ENGR. AZIZ							
			CHECKED BY:	ENGR. KAMRAN			**SCALE:**	**SHEET:** A3			
			APPROVED BY:	ENGR. KAMRAN							
			DATE:	OCTOBER, 2025							



Piller Foundation

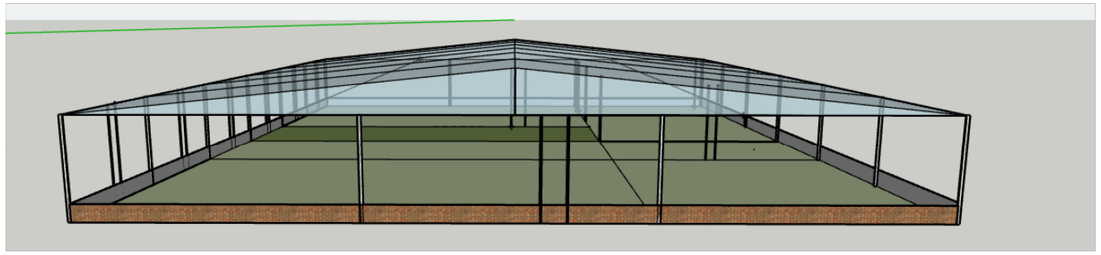


After Concreting

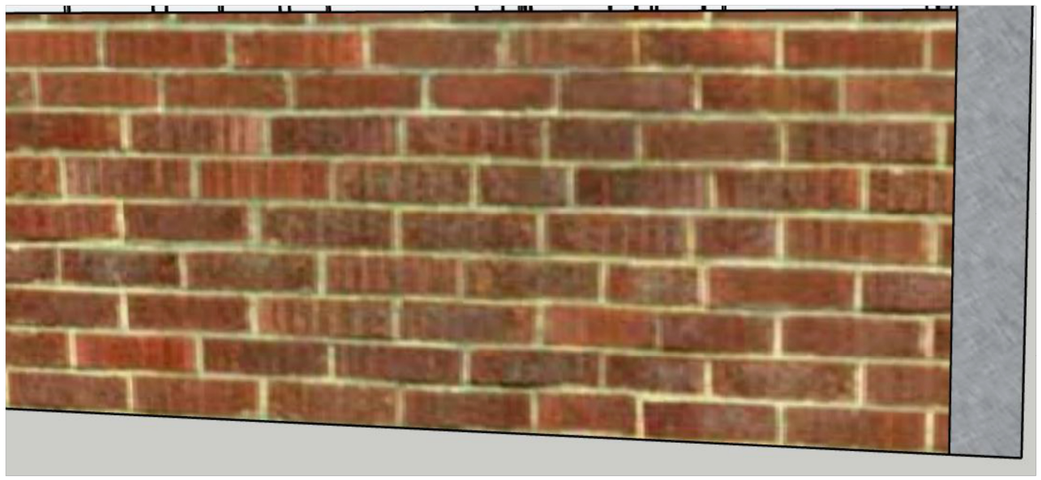
Engineering Consultancy Services Punjab(Pvt.)Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** PILLER FOUNDATION AND AFER CONCRETING	**DRAWING NO:** ARCH-07	
			DESIGNED BY:	ENGR. AZIZ			**SCALE:**	**SHEET:** A3
			CHECKED BY:	ENGR. KAMRAN				
			APPROVED BY:	ENGR. KAMRAN				
			DATE:	OCTOBER, 2025				

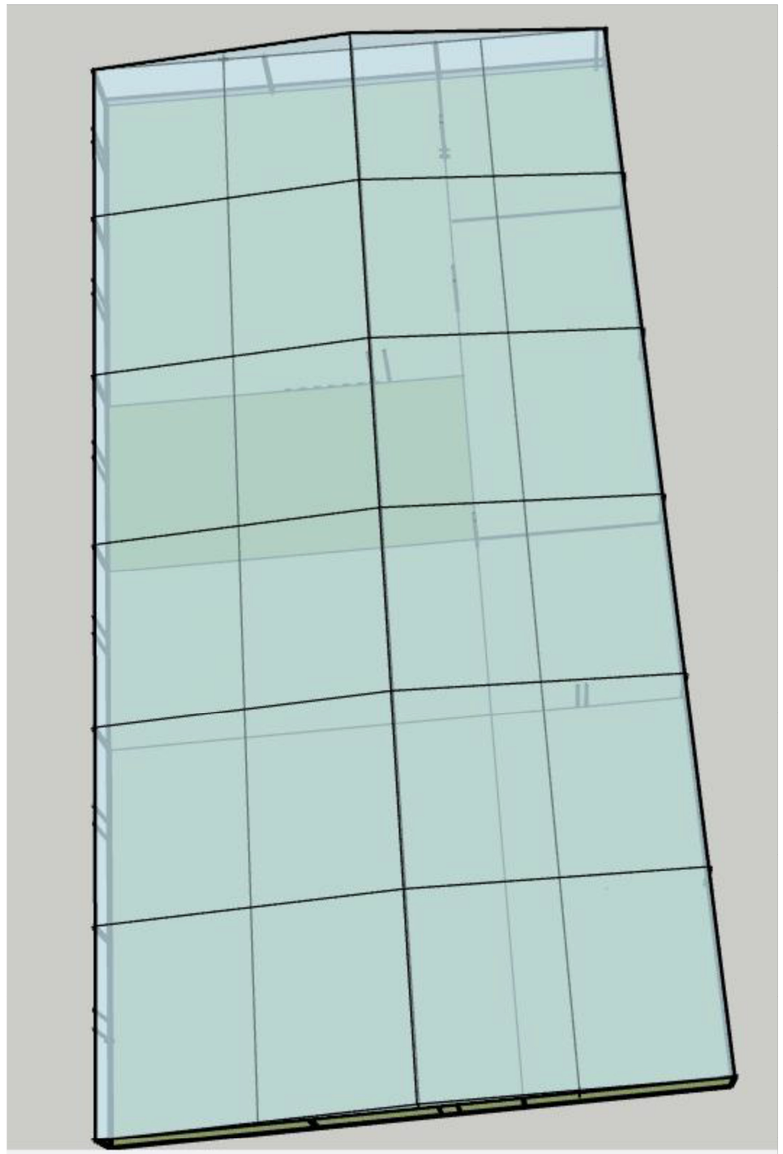
REV.	DESCRIPTION	SIGN.	DATE



Front View



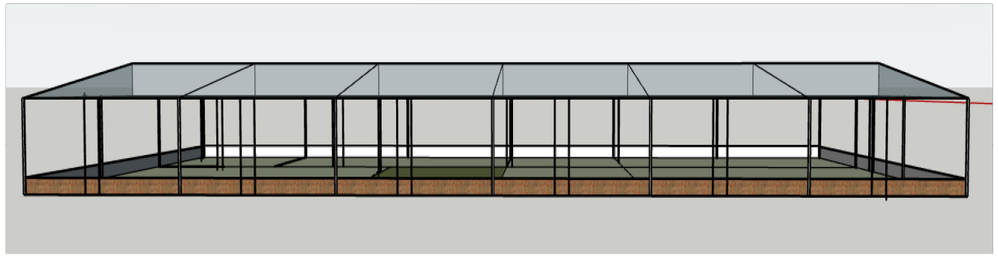
2’ Wall Around Structure



Top View

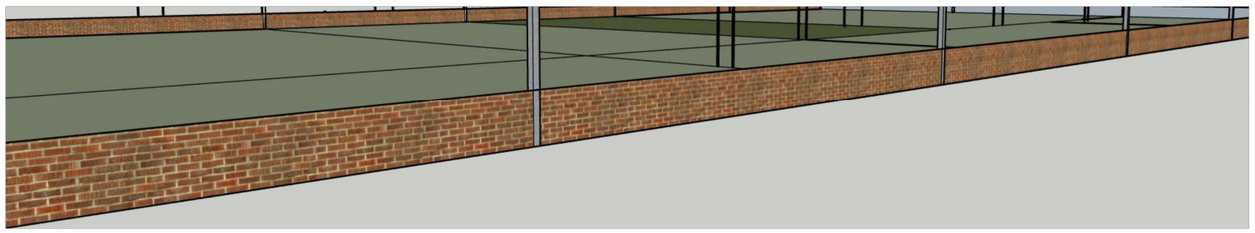
Engineering Consultancy Services Punjab (Pvt.) Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** 2' WALL AROUND STRUCTURE AND TOP VIEW	**DRAWING NO:** ARCH-08		REV.	DESCRIPTION	SIGN.	DATE
			DESIGNED BY:	ENGR. AZIZ			**SCALE:**	**SHEET:** A3				
			CHECKED BY:	ENGR. KAMRAN								
			APPROVED BY:	ENGR. KAMRAN								
			DATE:	OCTOBER, 2025								



Side View

Back Side View

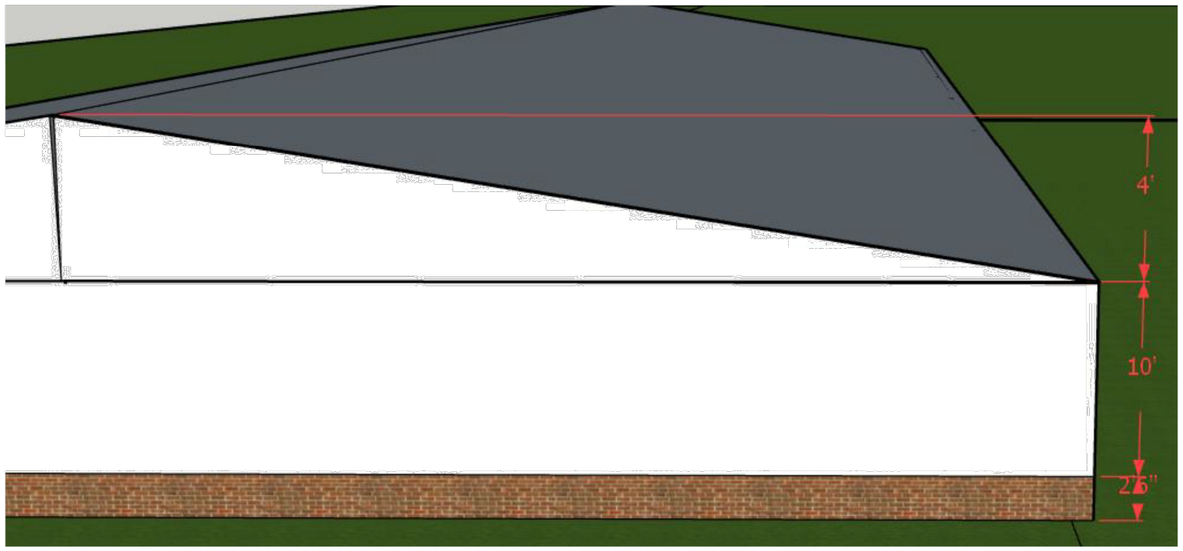


Side Wall

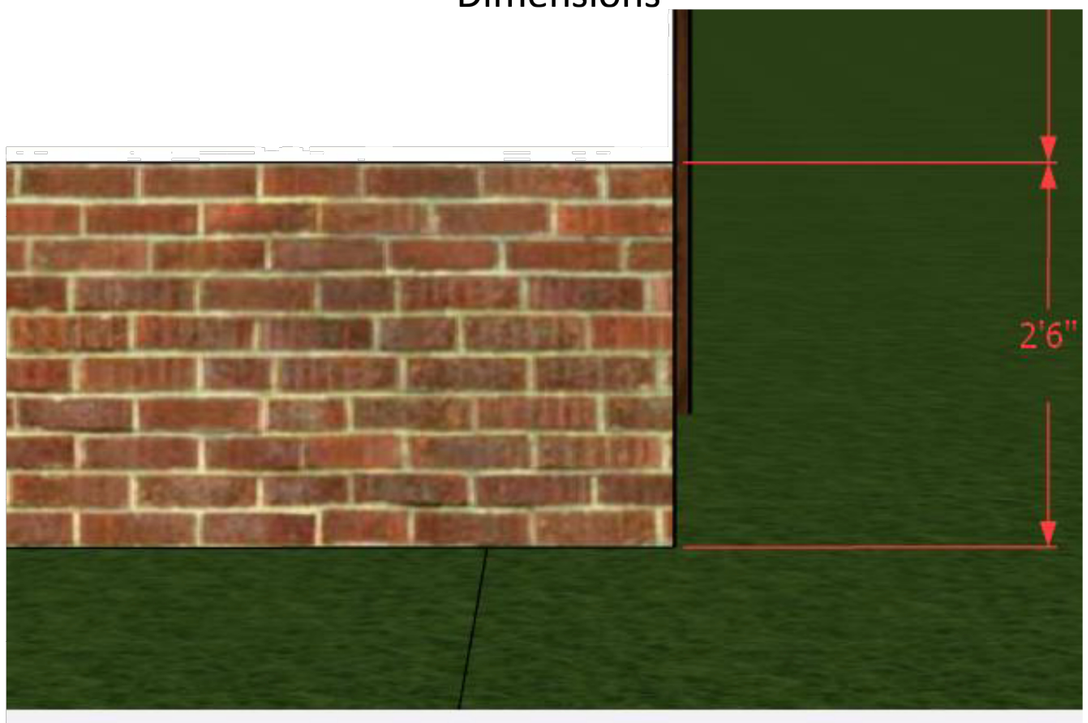
Engineering Consultancy Services Punjab (Pvt.) Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** SIDE VIEW, BACK SIDE VIEW AND SIDE WALL	**DRAWING NO:** ARCH-9	
			DESIGNED BY:	ENGR. AZIZ			**SCALE:**	**SHEET:** A3
			CHECKED BY:	ENGR. KAMRAN				
			APPROVED BY:	ENGR. KAMRAN				
			DATE:	OCTOBER, 2025				

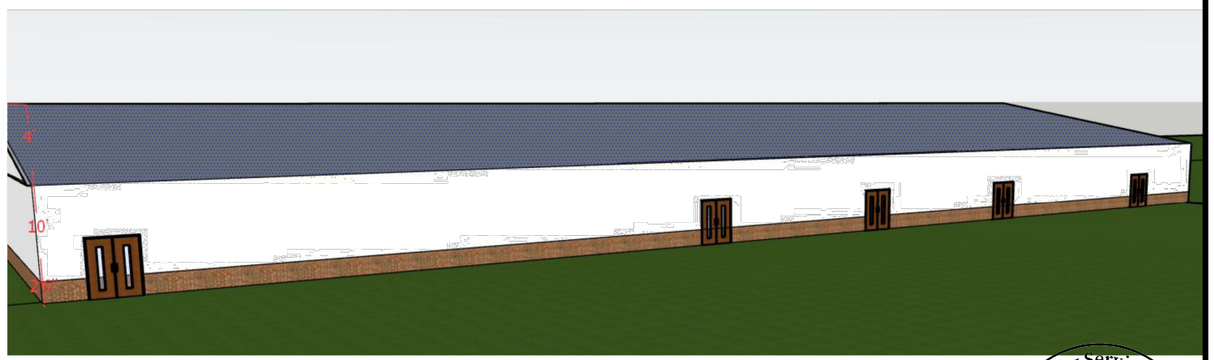
REV.	DESCRIPTION	SIGN.	DATE



Dimensions



Wall Height After floor



Side View

Engineering Consultancy Services Punjab(Pvt.) Ltd.
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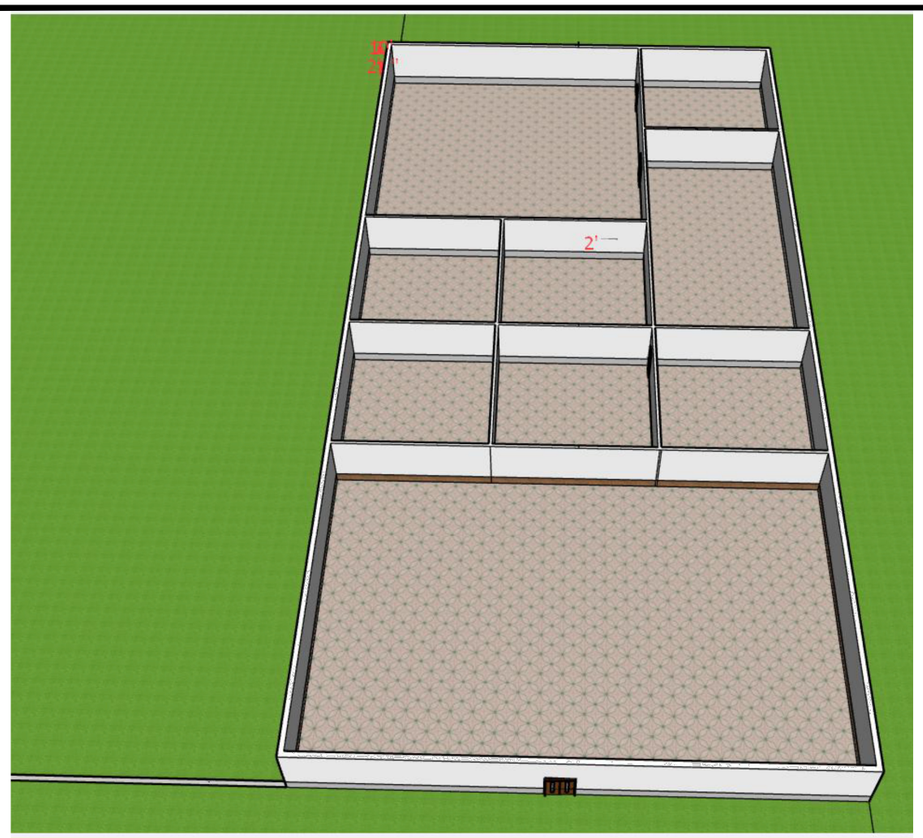
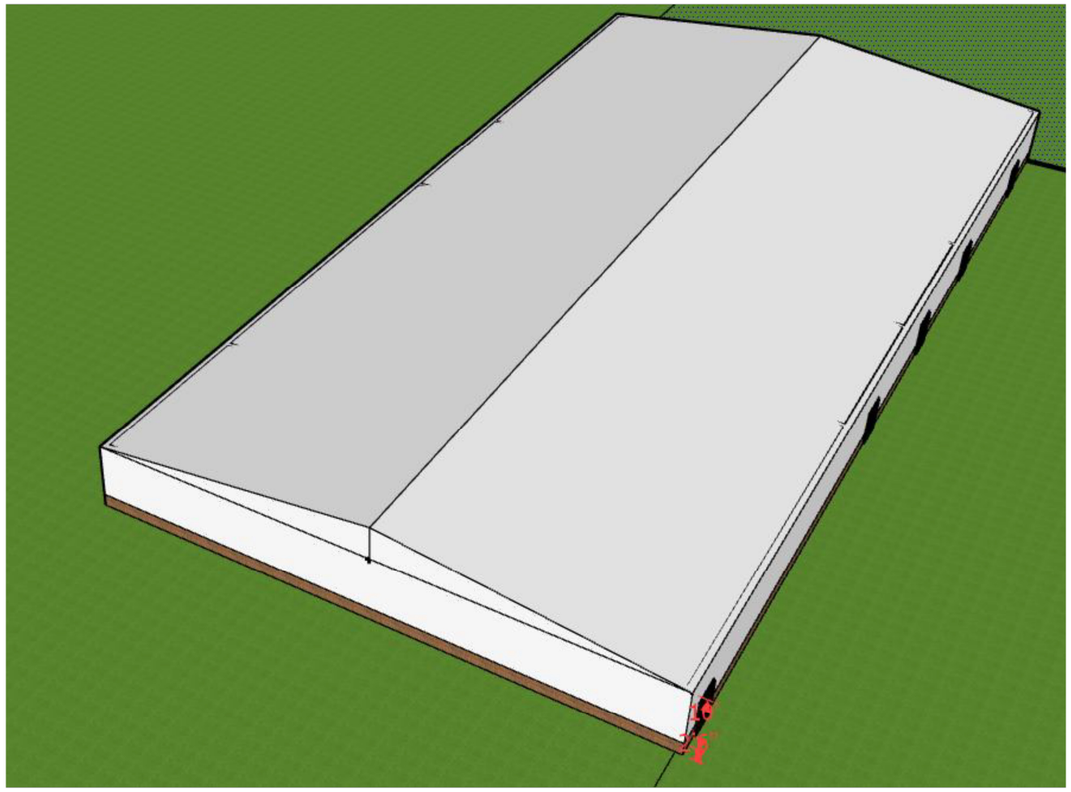
Wall Dimension

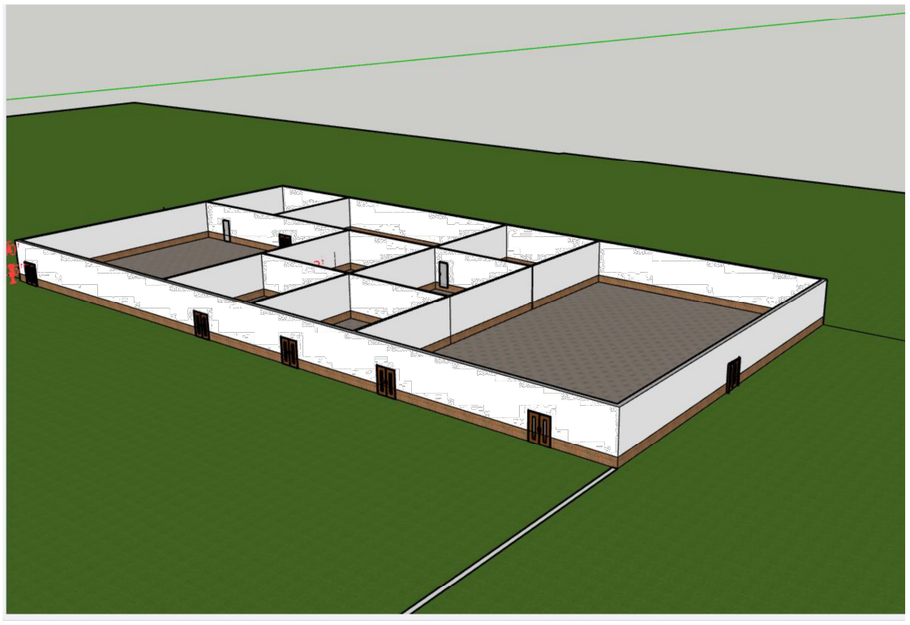
CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY: USMAN MUSTAFA DESIGNED BY: ENGR. AZIZ CHECKED BY: ENGR. KAMRAN APPROVED BY: ENGR. KAMRAN DATE: OCTOBER, 2025	**DRAWING TITLE:** DIMENSION DETAIL	**DRAWING NO:** ARCH-10 **SCALE:** **SHEET:** A3	REV. / DESCRIPTION / SIGN. / DATE



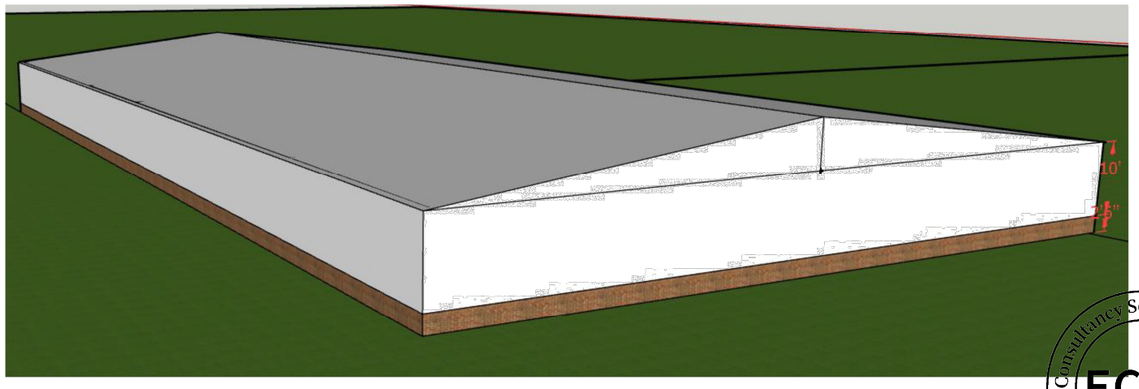
Inside View



Top View With Roof



Side View Without Roof



Complete Structure

Engineering Consultancy Services Punjab(Pvt.)Ltd. ECSP

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY:	USMAN MUSTAFA		**DRAWING TITLE:** INSIDE VIEW, SIDE VIEW WITHOUT ROOF, TOP VIEW WITH ROOF AND COMPLETE STRUCTURE	**DRAWING NO:** ARCH-11		
			DESIGNED BY:	ENGR. AZIZ			**SCALE:**	**SHEET:** A3
			CHECKED BY:	ENGR. KAMRAN				
			APPROVED BY:	ENGR. KAMRAN				
			DATE:	OCTOBER, 2025				

REV.	DESCRIPTION	SIGN.	DATE

1-GENERAL

1- READ ALL DRAWINGS IN CONJUNCTION WITH ARCHITECTURAL, PLUMBING/SEWERAGE, MECHANICAL, ELECTRICAL AND ANY OTHER RELEVANT DRAWINGS.

2- NOTES GIVEN IN THIS DRAWING ARE APPLICABLE TO ALL DRAWINGS UNLESS MENTIONED OTHERWISE. NOTES WRITTEN ON A DRAWING, SHALL BE APPLICABLE TO THAT PARTICULAR DRAWING ONLY UNLESS OTHERWISE CROSS-REFERRED.

3- ALL MATERIALS AND WORKMANSHIP SHALL CONFORM TO THE SPECIFICATIONS OF THE CONTRACT DOCUMENTS. IN ABSENCE OF ANY SPECIFICATIONS , ALL MATERIALS, TESTS AND WORKMANSHIP SHALL CONFORM TO RELEVANT ASTM, ACI/B.S CODES AND SHALL BE SUBJECT TO APPROVAL OF THE ENGINEER.

4- THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY OF THE STRUCTURES DURING CONSTRUCTION. HE SHALL ALSO VERIFY ALL DIMENSIONS AND LEVELS BEFORE EXECUTION OF WORK. ANY DISCREPANCY, ERROR OR OMISSION, IF FOUND, SHALL BE BROUGHT TO THE NOTICE OF THE ENGINEER FOR CORRECTION AND APPROVAL.

5- THE CONTRACTOR SHALL COORDINATE WITH ARCHITECTURAL AND ALL OTHER DICIPLINE DRAWINGS FOR SIZES AND LOCATION OF ALL STRUCTURAL MEMBERS, FLOORS, WALLS, OPENINGS, FLOOR FINISHES, PIPES ETC.

6- THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE EXECUTION OF DEWATERING SYSTEM WHERE SO REQUIRED DURING CONSTRUCTION. HE SHALL SUBMIT HIS METHODOLOGY FOR APPROVAL OF ENGINEER BEFORE ANY ACTIVITY.

7- THE CONTRACTOR SHALL SUBMIT SHOP DRAWINGS AND BAR BENDING SCHEDULES, 21 DAYS IN ADVANCE FOR APPROVAL, TO THE ENGINEER BEFORE PROCEEDING WITH THE WORK.

8-

9-

10- TERMITE CONTROL TREATMENT SHALL BE CARRIED OUT AS PER SPECIFICATIONS.

11- ALL LEVELS AND DIMENSIONS ARE IN MILIMETER UNLESS OTHERWISE NOTED.

12- DO NOT SCALE THE DRAWINGS WRITTEN DIMENSIONS, GIVEN ON THE DRAWING, SHALL GOVERN.

ALL FABRICATION, PAINTING, ERECTION AND QUALITY CONTROL IS TO BE DONE IN ACCORDANCE WITH THE LATEST APPLICABLE ACI/AISC/BRITISH SPECIFICATIONS.

BACKFILLING AROUND FOUNDATION SHOULD BE DONE SIMULTANEOUSLY ON BOTH SIDES TO AVOID ANY LATERAL DISPLACEMENT. BACKFILLING UNDER FLOORS AND AROUND FOUNDATIONS SHALL BE DONE IN LAYERS NOT EXCEEDING 6 iNCH. IN THICKNESS AND COMPACTED UPTO 95% MODIFIED PROCTOR DENSITY.

2-EXCAVATION

1- EXCAVATION SHALL BE MADE TO THE LINES AND LEVELS SHOWN ON THE DRAWINGS. THE ENGINEER MAY CHANGE THE EXTENT OR THE LEVELS OF EXCAVATION DURING THE PROGRESS OF WORKS.

2- THE CONTRACTOR MAY USE STABLE SLOPES OTHER THAN THOSE SHOWN ON THE DRAWINGS WITH THE PRIOR APPROVAL OF THE ENGINEER.

3- THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE STABILITY OF THE EXCAVATED AREA AND SHALL KEEP THE AREA IN A SOUND AND DRY CONDITION.

4- THE TERMITE CONTROL TREATMENT SHALL BE APPLIED AFTER EXCAVATION OF FOUNDATIONS ON BEDS AND SIDE SLOPES BEFORE CONCRETING OPERATION.

5- PART OF EXCAVATED MATERIAL, AS DIRECTED BY THE ENGINEER, SHALL BE PLACED IN STOCK PILES FOR BACKFILLING. ALL OTHER MATERIAL SHALL BE DISPOSED OF IN

6- APPROVED DISPOSAL AREA AS DIRECTED BY THE ENGINEER.

7- ANY OVER EXCAVATED AREAS OR DOUBTFUL PATCHES, AS DIRECTED BY THE ENGINEER

8- SHALL BE FILLED WITH LEAN CONCRETE.

3-FOUNDATIONS

01- THE DESIGN OF FOUNDATION IS BASED ON ASSUMED BEARING CAPACITY OF 0.75 TSF,

02- THE EXCAVATED BED OF FOUNDATION SHOULD BE THOROUGHLY COMPACTED PRIOR TO LAYING BLINDING CONCRETE IN FOUNDATION.

03- AS PER GEO-TECHNICAL INVESTIGATION REPORT OF THE PROJECT SITE. ANY SOFT SOIL / LOOSE FILL, IF ENCOUNTERED AT THE EXCAVATION BASE OF SHALLOW FOUNDATION SHOULD BE COMPLETELY REMOVED AND BACKFILLED WITH 2FT THICK WELL COMPACTED ENGINEERED FILL MATERIAL. THE ENGINEERED FILL SHOULD BE A-3 OR BETTER MATERIAL AS PER ASSHTO SOIL CLASSIFICATION. THE ENGINEERED FILL MATERIAL SHOULD BE PLACED IN LAYERS AND COMPACTED TO AT LEAST 95%OF MODIFIED PROCTOR DENSITY.

04- THE ENGINEERED FILL SHOULD BE GRANULAR MATERIAL HAVING % PASSING #200 SIEVE ≤ 15% WITH 1 ≤ Cc ≤ 3 AND Cu ≥ 4 AND SHOULD BE PLACED AND COMPACTED IN LAYERS OF 150 mm COMPACTED THICKNESS TO AT LEAST 95% OF MODIFIED AASHTO MAXIMUM DRY DENSITY.

05- SELECT SURPLUS SPOILS FROM THE EXCAVATED MATERIAL, NOT LARGER THAN 150 mm THICK COMPACTED LAYERS, SHALL BE USED FOR BACKFILLING OF FOUNDATION TRENCHES AND FOR GENERAL FILL. FILL MATERIAL SHALL BE FREE FROM CLODS, SALTS, SULPHATES AND ORGANIC OR OTHER FOREIGN MATERIAL. IN NO CASE, THE BACKFILL SOIL SHOULD BE EXPANSIVE IN NATURE. THE GENERAL FILL SHALL BE WELL COMPACTED IN LAYERS NOT EXCEEDING 200 mm IN THE DENSITY, 95 % OF MODIFIED AASHTO MAXIMUM DRY DENSITY. THE CONTRACTOR SHOULD SUBMIT HIS METHOD STATEMENT REGARDING PLACEMENT AND COMPACTION OF BACKFILLED & GENERAL FILL MATERIAL FOR APPROVAL FROM CLIENT / DESIGN ENGINEER.

06- THE FOUNDATION BED SOIL SHOULD BE PROTECTED FROM INGRESS OF WATER FROM ANY SOURCE BY PROVIDING ADEQUATE SURFACE DRAINAGE SYSTEM AND ENSURING LEAK PROOF JOINTINGS OF SEWERAGE AND WATER SUPPLY LINES. EXCESSIVE AND UNCONTROLLED WATERING FOR CURING AND COMPACTION OF BACKFILL MATERIAL SHOULD BE AVOIDED.

4-REINFORCED CONCRETE

01- STRUCTURAL DESIGN IS BASED ON THE NATIONAL BUILDING CODE OF PAKISTAN AND AMERICAN CONCRETE INSTITUTE CODE OF REINFORCED CONCRETE,ACI 318.

02- ALL STRUCTURAL CONCRETE SHALL CONFORM TO AMERICAN CONCRETE INSTITUTE (ACI) REQUIREMENTS.

03- THE MINIMUM WORK COMPRESSIVE CYLINDER STRENGTH OF LEAN/BLINDING CONCRETE SHALL BE 1000 psi AT 28 DAYS.

04- THE MINIMUM COMPRESSIVE CONCRETE CYLINDER STRENGTH OF STRUCTURAL CONCRETE FOR ALL STRUCTURAL MEMBERS SHALL BE 3000 psi AT 28 DAYS

05- NO CONCRETING SHALL BE CARRIED OUT UNTIL PERMISSION IS GIVEN IN WRITING.

5-REINFORCING STEEL

1- ALL REINFORCING STEEL SHALL BE DEFORMED BARS CONFORMING TO ASTM-A615 GRADE 60 HAVING A MINIMUM YIELD STRENGTH OF 60,000 psi.

2- WHEREVER POSSIBLE FULL LENGTH OF REINFORCING BARS SHALL BE USED TO AVOID UNNECESSARY LAPS AND WASTAGE.

3- CLEAR COVER TO REINFORCEMENT SHALL BE:

SRTUCT.ELEMENT	COVER (INCH)
FOOTINGS/RAFT	2"
BEAMS (BOTTOM)	1.5"
BEAMS (SIDES)	1"
COLUMNS	1.5"
SLABS	0.75"
WALLS (INNER FACE)	1"
WALLS (OUTER FACE IN CONTACT WITH EARTH)	2"

4- DEVELOPMENT LENGTH

fc' (PSI) \ Bar Size #	DEVELOPMENT LENGTH 'in' (Fy=60,000 PSI)					
	03	04	05	06	07	08
3000	25	33	42	50	73	83

5- SPLICE LENGTH

fc' (PSI) \ Bar Size #	SPLICE LENGTH 'in' (Fy=60,000 PSI)					
	03	04	05	06	07	08
3000	32	43	54	64	94	107

EXCEPT AS OTHERWISE SHOWN ON THE DRAWING, WHENEVER REINFORCING BARS OF DIFFERENT SIZES ARE TO BE LAP SPLICEED IN COMPRESSION, SPLICE LENGTH SHALL BE THE LARGER OF DEVELOPMENT LENGTH OF LARGER BAR, OR SPLICE LENGTH OF SMALLER BAR.

6- ALL REINFORCING STEEL SHALL BE ACCURATELY LOCATED IN THE FORMS AND HELD FIRMLY IN PLACE BEFORE AND DURING THE PLACEMENT OF CONCRETE, BY MEANS OF WIRE AND SUPPORTS ADEQUATE TO PREVENT DISPLACEMENT DURING THE COURSE OF CONSTRUCTION.

7- DISTRIBUTION REINFORCEMENT NOT SHOWN SHALL BE PROVIDED AS #3@9" c/c FOR 6 in THICK SLAB AND #3@7" c/c FOR 8 in THICK SLAB.

6-CONCRETE CONSTRUCTION

1- ALL STRUCTURAL SURFACES, AGAINST WHICH EARTH IS TO BE FILLED, SHALL BE COATED WITH TWO COATS OF HOT BITUMEN OF 10/20 GRADE AT THE RATE OF 0.4 lb/ft^2/coat.

2- THE CONTRACTOR SHALL SUBMIT CONCRETE POURING SCHEDULE FOR ENGINEER'S APPROVAL. NO CONCRETE SHALL BE PLACED UNTIL WRITTEN PERMISSION IS GIVEN BY THE ENGINEER.

3- DURING CONSTRUCTION, STACKING OF CONSTRUCTION MATERIALS, BRICKS, etc. SHOULD BE AVOIDED ON SLAB PANELS.

4- BEFORE CASTING OF ANY STRUCTURAL MEMBER, THE CONTRACTOR SHALL ENSURE THAT ALL EMBEDDED ITEMS FOR ELECTRICAL, MECHANICAL, HVAC, PLUMBING, STRUCTURAL STEEL AND OTHER WORKS ARE PROPERLY LOCATED AND FIRMLY SECURED IN PLACE.

7-CONSTRUCTION JOINTS

1- JOINTS, NOT SHOWN ON THE DRAWINGS, SHALL BE SO MADE AND LOCATED AS TO APPROVAL OF THE ENGINEER. THEY SHALL BE LOCATED WITHIN THE MIDDLE THIRD OF THE LEAST IMPAIR THE STRENGTH OF THE STRUCTURE AND SHALL NEED PRIOR SPANS OF SLABS, BEAMS AND JOISTS. IN WALLS AND COLUMNS SHALL BE AT THE UNDER-SIDE OF FLOOR, SLABS OR BEAMS AND AT THE TOP OF FOOTINGS OR FLOOR SLABS.

2- JOINTS SHALL BE PERPENDICULAR TO MAIN REINFORCEMENT. ALL REINFORCING STEEL SHALL BE CONTINUED ACROSS JOINTS.

3- BEAMS, GIRDERS AND HAUNCHES SHALL BE PLACED MONOLITHICALLY AS PART OF SLAB SYSTEM UNLESS OTHERWISE SHOWN IN DESIGN DRAWINGS OR SPECIFICATIONS.

8-SHORING AND BRACING

1- SHORE AND BRACE ALL PARTS OF THE BUILDING DURING CONSTRUCTION, TO THE EXTENT NECESSARY TO ENSURE COMPLETE SAFETY, STRENGTH AND SERVICEABILITY OF ALL STRUCTURAL ELEMENTS UNDER ALL CONDITIONS OF LOADS, WHICH MAY OCCUR DURING CONSTRUCTION. SUCH SHORING AND BRACING IS THE CONTRACTOR'S SOLE RESPONSIBILITY AND IS NOT SHOWN ON STRUCTURAL DRAWINGS OR SPECIFIED IN THE PROJECT SPECIFICATIONS.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY AND STABILITY OF THE ADJACENT EXISTING STRUCTURES AND SHALL PROVIDE ADEQUATE

2- SHORING AND BRACING DURING EXCAVATION AND CONSTRUCTION, WHEREVER AND WHENEVER REQUIRED TO THE SATISFACTION OF THE ENGINEER.

9-BRICK WORK

1- ALL BRICK WORK SHALL CONFORM TO SPECIFICATIONS GIVEN IN THE CONTRACT DOCUMENT.

2- ALL BRICKS SHALL BE SOUND, HARD, WELL BURNT AND OF UNIFORM SIZE,COLOUR AND TEXTURE. DIMENSIONAL VARIATION IN SIZES SHALL NOT EXCEED 0.12in.

3- ALL BRICK WORK SHALL BE ERECTED PLUMB AND TRUE TO LINE AND LEVEL. THE MAXIMUM VARIATION IN ANY STOREY HEIGHT OR ANY LENGTH OF WALL SHALL BE 0.12in IN 10ft.

4- ALL BRICK MASONRY FOR WALLS SHALL BE REINFORCED WITH MIILD STEEL FLAT/REINFORCEMENT AS SHOWN ON THE DRAWINGS.

5- TESTING OF COMPRESSIVE STRENGTH OF MASONRY SHALL BE DONE, IF SO DIRECTED BY THE ENGINEER, IN ACCORDANCE WITH ASTM E447 OR PAKISTAN STANDARD-208.

6- THE AVERAGE COMPRESSIVE STRENGTH OF TWELVE BRICKS SHALL BE NOT LESS THAN 2000 psi AND SHALL BE NOT LESS THAN 1500 psi FOR ANY INDIVIDUAL BRICK.

7- BRICK SHALL BE LAID "FROG" UPWARD WITH MORTAR JOINT AND IN ENGLISH AND FLEMISH BOND OR AS DIRECTED BY THE ENGINEER.

8- ALL BRICK MASONRY WORK FOR SUB STRUCTURE SHALL BE CARRIED OUT BY USING 1:5 CEMENT SAND MORTAR

9- ALL BRICK MASONRY WORK FOR SUPER STRUCTURE SHALL BE CARRIED OUT BY USING 1:4 CEMENT SAND MORTAR

10- ALL BRICK MASONRY WORK FOR BOUNDARY WALL SHALL BE CARRIED OUT BY USING 1:3 CEMENT SAND MORTAR

11- ALL BRICK MASONRY WALLS SHALL BE PLASTERED WITH 1:4 CEMENT SAND MORTAR.

10-ELECTRICAL CONDUITS

1- CONDUITS, FOR ELECTRICAL WORKS, SHALL BE PLACED WITHIN THE REINFORCED CONCRETE. THEY SHALL BE PLACED WITHIN THE MIDDLE THIRD OF THE SECTION IN BEAMS AND WITHIN THE MIDDLE HALF OF THE THICKNESS IN SLABS. SPACING BETWEEN PARALLEL CONDUITS SHALL NOT BE LESS THAN 6 in.

11- STRUCTURAL STEEL

1- FABRICATION, PAINTING, ERECTION AND QUALITY CONTROL IS TO BE PERFORMED IN ACCORDANCE WITH AISC SPECIFICATIONS.

2- ALL STRUCTURAL STEEL SHALL CONFORM TO THE REQUIREMENTS OF ASTM A-36 OR EQUIVALENT AS APPROVED BY THE ENGINEER.

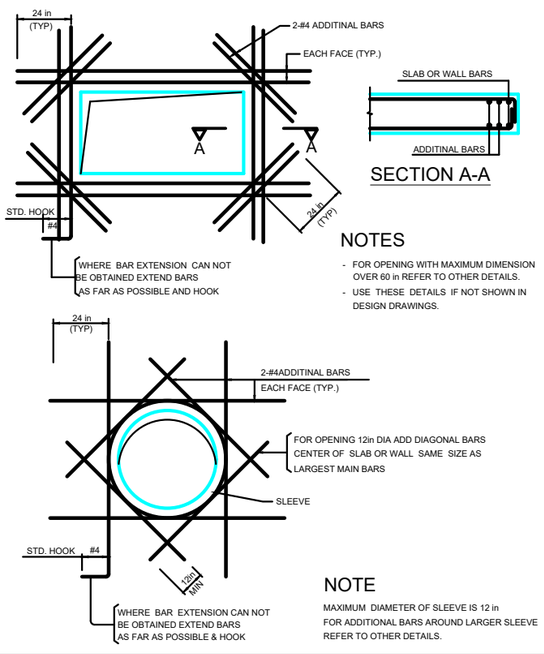
3- ALL WELDING SHALL BE DONE IN ACCORDANCE WITH THE REQUIREMENTS OF AMERICAN WELDING SOCIETY, AWS, SPECIFICATIONS USING ELECTRODES E70XX.

4- THE ANCHOR BOLTS SHALL CONFORM TO ASTM A325, A490, F1554 AND A307 RESPECTIVELY.

5- THE CONTRACTOR SHALL SUBMIT SHOP DRAWINGS BEFORE EXECUTION OF STEEL STRUCTURAL WORKS.

6- THE CONTRACTOR SHALL SUBMIT PEB DESIGN AND DRAWING FOR THE BUILDINGS CONTAINING STEEL STRUCTURES FOR CONSULTANT'S APPROVAL.

12- REINFORCEMENT DETAIL AT OPENING IN SLAB/SLEEVES.



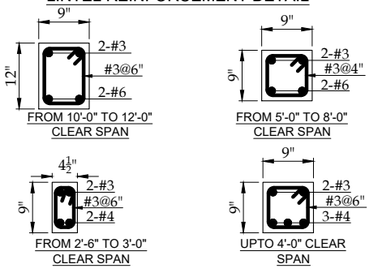
NOTES

- FOR OPENING WITH MAXIMUM DIMENSION OVER 60 in REFER TO OTHER DETAILS.
- USE THESE DETAILS IF NOT SHOWN IN DESIGN DRAWINGS.

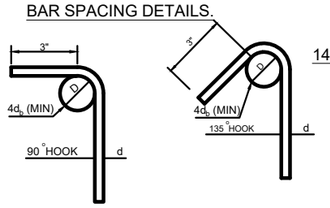
NOTE

MAXIMUM DIAMETER OF SLEEVE IS 12 in
FOR ADDITIONAL BARS AROUND LARGER SLEEVE
REFER TO OTHER DETAILS.

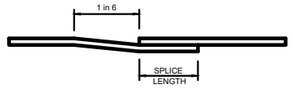
LINTEL REINFORCEMENT DETAIL



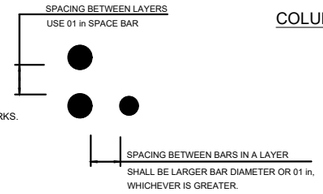
14- STIRRUPS, HOOKS, SPLICES AND REINFORCEMENT BAR SPACING DETAILS.



STIRRUPS AND TIE HOOKS

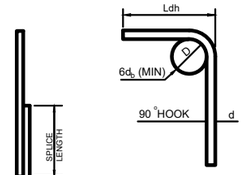


BEAM SPLICE (TYP.)

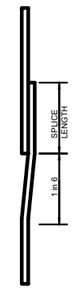


BAR SPACING IN BEAM

14a- STANDARD HOOK FOR BEAM & COLUMN BARS



HOOKS FOR DEFORMED BARS IN TENSION

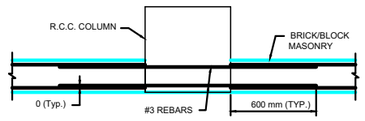


COLUMN SPLICE (TYP.)

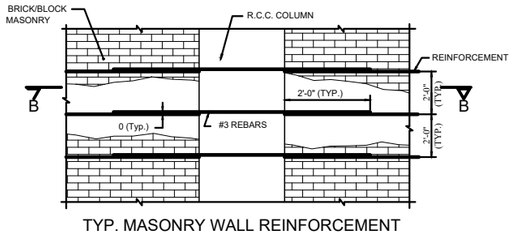
VERTICAL BAR SPACING IN COLUMNS

BAR NO.	Ldh
#4	8"
#6	12"
#8	16"

15- BRICK MASONRY REINFORCEMENT DETAIL.



PLAN B-B



TYP. MASONRY WALL REINFORCEMENT

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB

DRAWN BY:	SHOAIB	
DESIGNED BY:	ENGR. ASAD	
CHECKED BY:	ENGR. SABIR	
APPROVED BY:	ENGR. SABIR	
DATE:	OCTOBER, 2025	

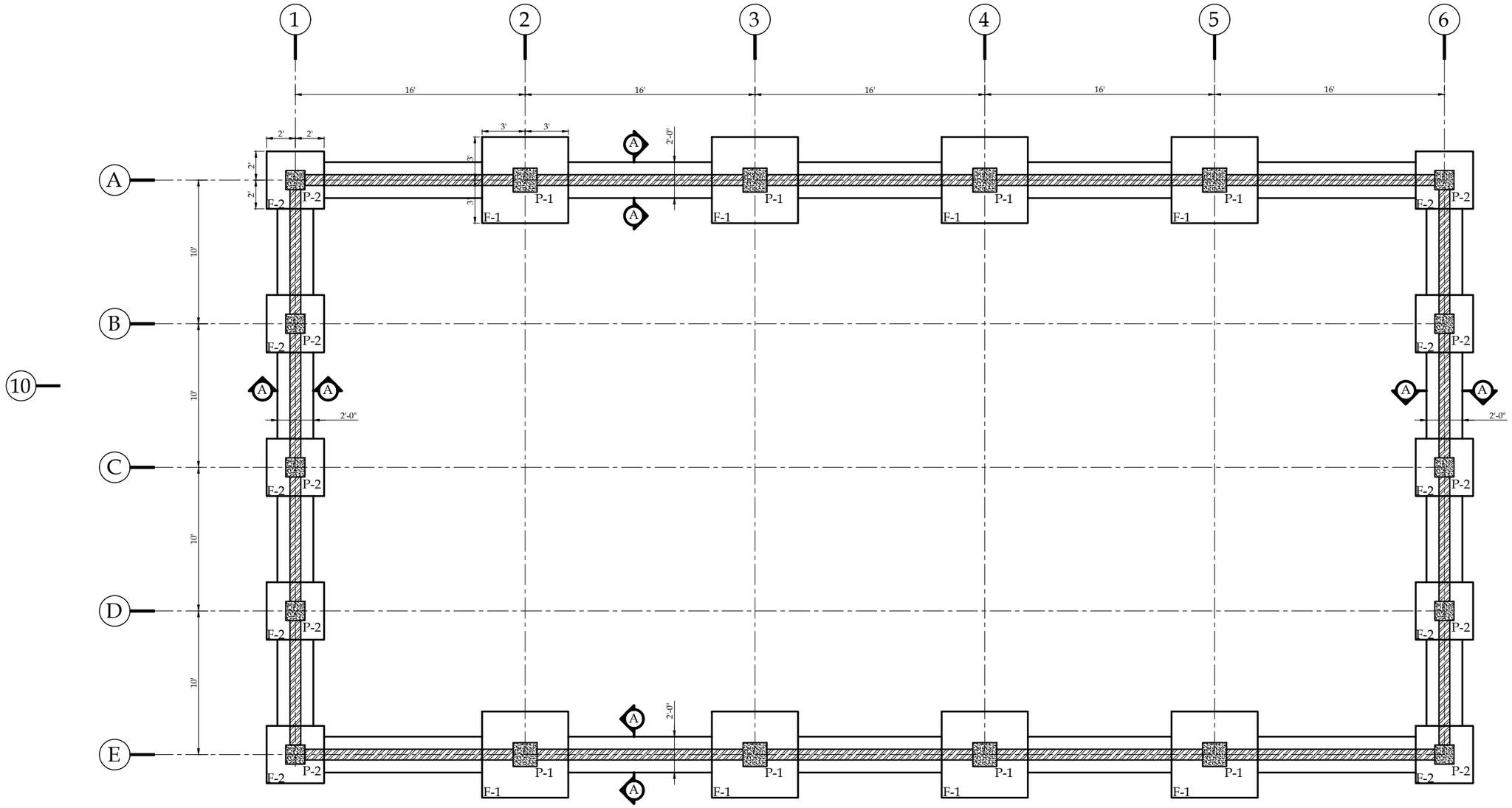
DRAWING TITLE: GENERAL NOTES

DRAWING NO: STR-01

SCALE:

SHEET: A3

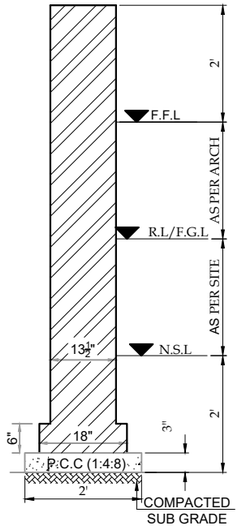
REV.	DESCRIPTION	SIGN.	DATE



FOUNDATION LAYOUT PLAN

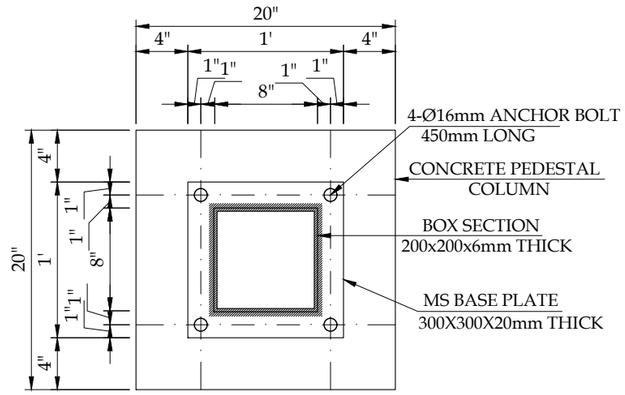
SCALE:- 1/8" = 1'-0"

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY: SHOAIB DESIGNED BY: ENGR. ASAD CHECKED BY: ENGR. SABIR APPROVED BY: ENGR. SABIR DATE: OCTOBER, 2025	**DRAWING TITLE:** FOUNDATION & PLINTH BEAM LAYOUT PLAN	**DRAWING NO:** STR-02 **SCALE:** **SHEET:** A3	REV. / DESCRIPTION / SIGN. / DATE	



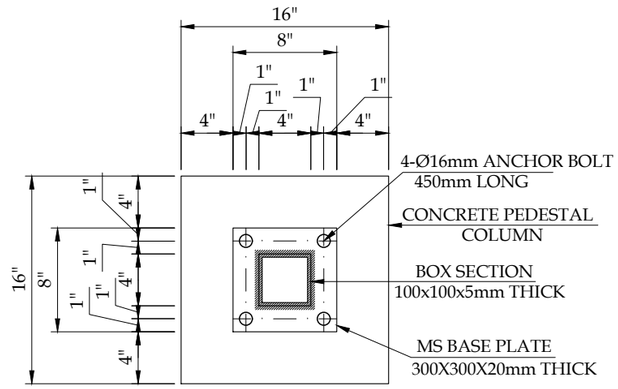
TOE WALL SECTION A-A

SCALE:- 3/8" = 1'-0"



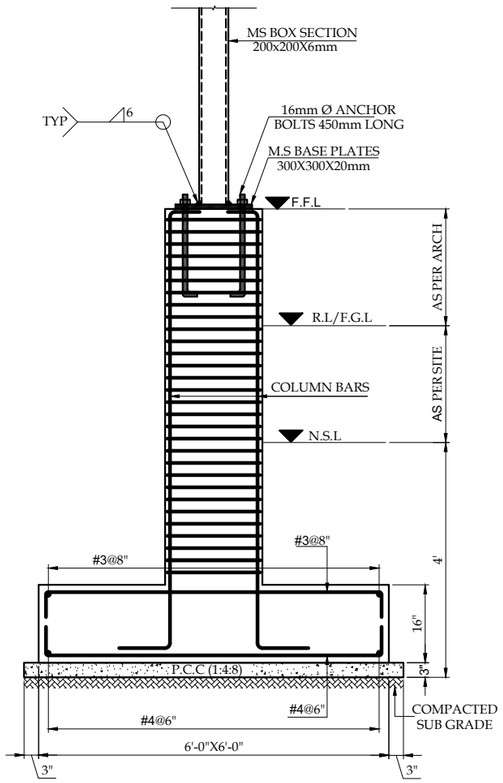
MS BASE PLATE FOR FOOTING F-1

SCALE:- 1" = 1'-0"



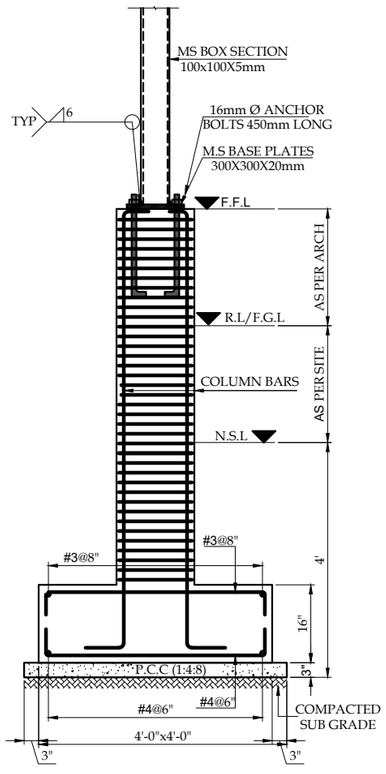
MS BASE PLATE FOR FOOTING F-2

SCALE:- 1" = 1'-0"



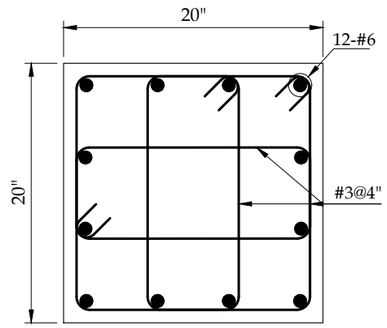
FOOTINING F-1

SCALE:- 3/8" = 1'-0"



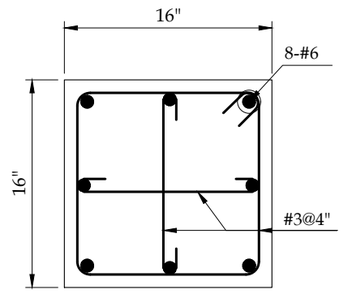
FOOTINING F-2

SCALE:- 3/8" = 1'-0"



PEDESTAL P-1 FOR FOOTING F-1

SCALE:- 1" = 1'-0"



PEDESTAL P-2 FOR FOOTING F-2

SCALE:- 1" = 1'-0"

CLIENT: PUNJAB SAAF PANI AUTHORITY	**CONSULTANTS:** ENGINEERING CONSULTANCY SERVICES PUNJAB (PVT) LTD (ECSP). 83-A, E/1, Main Boulevard, Gulberg III, Lahore, Pakistan. +92-42-99333976, +92-42-5717681-4 info@ecsp.com.pk www.ecsp.com.pk	**PROJECT:** SUPPLY, INSTALLATION & COMMISSIONING OF WATER BOTTLING PLANT (5000 LPH) FOR PROVISION OF SAFE DRINKING WATER IN DISTRICT BAHAWALPUR, RAHIM YAR KHAN, DERA GHAZI KHAN AND KHUSHAB	DRAWN BY: SHOAIB DESIGNED BY: ENGR. ASAD CHECKED BY: ENGR. SABIR APPROVED BY: ENGR. SABIR DATE: OCTOBER, 2025	**DRAWING TITLE:** FOUNDATION SECTIONAL DETAILS	**DRAWING NO:** STR-03 **SCALE:** **SHEET:** A3	REV. / DESCRIPTION / SIGN. / DATE		